Project Option Trading — ML Research Review & Handoff

Sep 27, 2026 · @Tony

Overview

The conversation concludes that no single ML model wins in options trading; a layered system that predicts volatility, selects a strategy by regime, sizes and hedges positions, and trades only when the edge beats costs is the recommended design.

Source: the shared ChatGPT project chat "Project Option trading" (link). It contains four research exchanges plus a final request for a handoff file. This document summarizes each exchange, pulls out the decisions, and adds a critical review.

#	Question asked	Main output
1	Top 20 profitable ML models, with evidence, and the factors that make an ML trading model succeed	Ranked list of 20 models, best models per task, 20 ranked success factors, a reference architecture
2	Summary table of all 20 models, with suitability for intraday, daily, medium-term and long-term trading	Model × horizon matrix; six core model families selected
3	Three best option-trading frameworks plus two institutional frameworks (#4, #5), with goals and rationale	Five-framework comparison; objective hierarchy; six-layer final architecture
4	Add horizon suitability to the five-framework table	Framework × horizon table; daily + medium-term chosen as primary horizon

The chat references an earlier study on the "top 20 option strategies" that is not part of this shared link. The earlier handoff file ChatGPT produced was also flagged by ChatGPT itself as incomplete.

Discussion 1 — Top 20 ML models for options

XGBoost ranked #1 overall, but the key finding was that the best model is the one with a stable, after-cost, out-of-sample edge, not the lowest prediction error. A cited 2026 study on 18.2 million minute-bar S&P 500 call observations found lower pricing error did not translate into better trading P&L.

Rank	Model	Best use in options	Key evidence cited	Rating
1	XGBoost / gradient-boosted trees	IV, returns, probability of profit, trade selection	Boosting beat Black-Scholes / Corrado-Su in option pricing; XGBoost and RF most accurate across regimes	5

Rank	Model	Best use in options	Key evidence cited	Rating
2	LSTM / Bi-LSTM	IV/RV forecasting, sequential signals	2026 equity-option study (2018–2023): LSTM MSE ~4.35%, R^2 ~53%; pooled "universal" models beat per-stock models	5
3	Transformer / Autoformer	Long-horizon, multi-source forecasting	Autoformer beat simpler NNs on all US stocks, 920 features, data from 1957	5
4	LightGBM	IV surface, very large tabular data	VAE → LightGBM → Monte Carlo beat Black-Scholes on NIFTY50 options	5
5	Temporal Fusion Transformer	Multi-horizon, heterogeneous features	Built for multi-horizon forecasting with interpretable attention	5
6	GRU	Hedging, IV/RV	GRU-based hedging model beat benchmarks on S&P 500 options (J. Banking & Finance)	4.5
7	Random Forest	Robust baseline, classification	Strong in option pricing, but worst model in an FX-option IV study	4.5
8	MLP	Cross-sectional pricing / returns	Gu–Kelly–Xiu: NN timing Sharpe 0.77 vs 0.51 buy-and-hold; long-short Sharpe 1.35; best at ~3 layers	4.5
9	Graph Neural Network	Whole option chain / surface	GNN with momentum spillover improved RMSE ≥8.81% on SSE 50 ETF options	4.5
10	PPO (deep RL)	Dynamic hedging and trading	Among top performers in an 8-algorithm RL hedging comparison	4.5
11	DDPG / TD3	Continuous hedge sizing	Beat Black-Scholes delta hedge under costs for American options	4
12	CNN	IV surface as a 2D image	Predicted IV surface on SSE 50 ETF options, beat backprop NNs	4
13	VAE	Distribution / surface compression	Reproduces skew, kurtosis, fat tails in the NIFTY50 framework	4
14	Neural SDE	Pricing, scenario generation	Compared with Black-Scholes, Dupire and Heston on real options	4
15	Gaussian Process / Deep Kernel	Small-data pricing with uncertainty	DKL beat Least-Squares Monte Carlo for high-dim American options	4

Rank	Model	Best use in options	Key evidence cited	Rating
16	N-BEATS / N-BEATS-GAN	Time-series forecasting + risk	Limited option-specific evidence	3.5
17	Elastic Net	Benchmark, feature selection	Positive out-of-sample R^2 where OLS failed (Gu–Kelly–Xiu)	3.5
18	SVR	Small/medium nonlinear data	Used as benchmark in pricing and volatility studies	3.5
19	CatBoost	Categorical / mixed features	ML beat Black-Scholes/Heston in several configurations	3.5
20	Ensemble / stacking	Production robustness	DNN + GBT + RF combination was strongest in S&P 500 stat-arb study	5*

Seven possible prediction targets were distinguished: direction, return, realized volatility, implied volatility, option fair value, probability of a profitable trade, and optimal action (the RL problem). Choosing the target was called the most important decision in the project.

Best models by task (top three of each list):

Task	1st	2nd	3rd
Realized volatility	LSTM	XGBoost	LightGBM
Implied volatility	LSTM / Bi-LSTM	XGBoost	LightGBM
Option pricing	XGBoost	LightGBM	MLP
Stock direction	XGBoost	MLP	Random Forest
Dynamic hedging	PPO	Policy-gradient RL	DDPG / TD3

Recurring lesson: simple models often win with limited data; complex models only pay off as data quality and quantity grow. Finance has low signal-to-noise, nonstationarity and changing regimes.

Discussion 1b — What makes an ML trading model succeed

Data quality, leakage prevention, target definition, transaction-cost realism and out-of-sample validation were all rated 10/10 — judged more important than the choice between XGBoost and LSTM.

Rank	Factor	Score	Rank	Factor	Score
1	Data quality	10	11	Model calibration	9.5
2	Leakage prevention	10	12	Objective / loss function	9.5
3	Target definition	10	13	Ensemble diversity	9.4
4	Transaction-cost realism	10	14	Interpretability	9.0

Rank	Factor	Score	Rank	Factor	Score
5	Out-of-sample validation	10	15	Computational efficiency	8.5
6	Regime robustness	9.9	16	Hyperparameter optimization	8.5
7	Feature quality	9.8	17	Retraining frequency	8.5
8	Position sizing	9.8	18	Uncertainty estimation	8.5
9	Tail-risk management	9.8	19	Execution infrastructure	8.5
10	Nonstationarity handling	9.7	20	Model governance	8.0

Key points behind the factors

- **Leakage:** option data is especially prone to it — expiring contracts, IV derived from prices, asynchronous bid/ask, corporate actions, survivorship bias.
- **Targets:** predict future IV, IV change, mispricing, future RV, expected trade P&L or P(P&L > 0) rather than raw next-day option price. The last two are most useful for trading.
- **Costs:** a model forecasting +0.20% per trade is worthless at 0.35% cost. A cited walk-forward study found XGBoost, LSTM and iTransformer signals failed after a 10 bp cost until a cost-aware threshold was added.
- **Metrics:** track prediction metrics (RMSE, MAE, R^2, log loss, Brier, calibration) separately from trading metrics (net Sharpe, Sortino, Calmar, max drawdown, expected shortfall, turnover, profit factor).
- **Features:** ten families — price, volatility, options (ATM IV, 25Δ skew, term structure, IV-RV spread), Greeks, flow, liquidity, VIX complex, macro, sentiment. Gu–Kelly–Xiu found momentum, liquidity and volatility interactions most predictive.
- **Universal models:** one pooled model across all underlyings (with per-underlying embeddings) beat separate per-stock models.
- **Surface modelling:** treat the whole IV(K, T) surface as one observation (CNN for local patterns, GNN for contract links, VAE for compression).
- **Calibration and distributions:** a calibrated 70% probability is worth more than a raw point forecast; prefer predicting the distribution of returns. A +0.6% forecast at 92% confidence was preferred over +1% at 51%.
- **LLMs:** not for numeric prediction; use them to extract signals from news, earnings calls, SEC filings and central-bank language (consistent with IMF and BIS views).
- **Institutional caution:** the Bank of England notes firms mainly use autonomous AI for research, coding and surveillance, not autonomous trading. The IMF warns of correlated behaviour and crowding.

Validation protocol agreed

1. Statistical layer — does it predict IV, RV, returns?
2. Economic layer — positive expectancy, Sharpe, Sortino, drawdown?
3. Execution layer — still profitable after spread, commission, slippage, impact, latency?

4. Crisis test — 2008, 2020, 2022 and gap events separately.

Walk-forward example: train 2010–2018, validate 2019–2020, test 2021; then roll forward one year. Never random 80/20 splits.

Explicit "do not" list: LSTM on closing prices alone; trading whenever predicted price > market; accuracy as the main metric; optimizing Sharpe on one period; 500 unregularized indicators; ignoring bid/ask; one model forever.

Learning-capability criteria (ranked): nonlinearity, temporal memory, cross-sectional learning, regime adaptation, transfer learning, uncertainty estimation, online adaptation, robustness to missing data, interpretability, retraining speed.

Discussion 2 — Which model suits which trading horizon

Daily trading was named the largest opportunity: enough observations for ML without the extreme microstructure costs of intraday. Horizons mean the prediction/rebalancing horizon, not a promise of profit.

Star ratings from ChatGPT's final model × horizon matrix (5 = best). Medium term = 1–12 weeks; long term = 3–12+ months.

Model	Why it is good	Intraday	Daily	Medium	Long	Best role
XGBoost	Nonlinear interactions on tabular data, fast	5	5	5	3	IV/RV, trade selection, mispricing
LSTM / Bi-LSTM	Volatility persistence, sequences	4.5	5	5	3	IV, RV, option-price forecasts
LightGBM	Scales to huge option datasets	5	5	4.5	3	IV surface, cross-sectional ranking
Transformer / Autoformer	Long-range, many features	3	5	5	5	Multivariate forecasting
Random Forest	Robust, low tuning	4	4.5	4	3	Classification, regime, filtering
TFT	Several horizons at once	3	5	5	5	Multi-horizon IV/RV
GRU	Lighter LSTM	4.5	5	4.5	3	Hedging, IV/RV
MLP	Simple, strong with engineered features	4	4.5	4.5	4	Cross-sectional pricing
GNN	Links across strikes and	3	4.5	5	5	Option chain /

Model	Why it is good	Intraday	Daily	Medium	Long	Best role
	maturities					surface
CNN	IV surface as an image	4	4.5	4	3	Smile, skew, term structure
PPO	Learns decisions, not forecasts	4	4.5	4	3	Sizing, hedging, execution
DDPG / TD3	Continuous actions	4	4.5	4	3	Dynamic hedging
Neural SDE	Learns the price process	2	3	4	5	Pricing, simulation
VAE	Latent factors, distributions	3	4	5	5	Surface compression
Gaussian Process / DKL	Uncertainty estimates	2	4	4.5	4.5	Pricing + confidence
N-BEATS	Dedicated time-series model	3	4	4.5	4	Volatility / returns
Elastic Net	Interpretable benchmark	5	5	4	4	Baseline, feature selection
SVR	Small/medium nonlinear data	3	4	4	3	Pricing / IV
CatBoost	Categorical features	5	5	4	3	Cross-sectional prediction
Ensemble / stacking	Offsets individual weaknesses	5	5	5	5	Final trade-ranking layer

Horizon short-lists

- **Intraday:** XGBoost, LightGBM, LSTM, GRU, RL — speed matters; a Transformer may predict better yet lose money on turnover.
- **Daily:** XGBoost, LSTM, LightGBM, Transformer, GNN.
- **Medium term:** LSTM, Transformer, TFT, GNN, XGBoost — time to exploit IV/RV divergence, regimes, term structure, earnings.
- **Long term:** Transformer, TFT, GNN, VAE, Neural SDE, simpler regularized models — few independent observations, so big deep models can underperform.

Corrected reading of the ranking: XGBoost wins general-purpose; LSTM wins sequential volatility; Transformers win large multi-source data; GNN wins option-chain structure; RL wins hedging decisions; Elastic Net is the benchmark.

Decision: narrow the experiment to six core families — XGBoost, LSTM, Transformer/TFT, GNN, PPO/deep hedging, Elastic Net — and keep the other 14 as secondary benchmarks. Separate the prediction layer from the decision layer.

Discussion 3 — Five frameworks, their goals and rationale

The Adaptive ML Volatility + Relative-Value framework ranked #1, but the final recommendation was to combine frameworks 1–3 and use 4 and 5 as institutional benchmarks. Frameworks 4 and 5 are public institutional archetypes, not proprietary bank systems.

#	Framework	Core philosophy	Main goal	Profit	Risk control	Why it was chosen
1	Adaptive ML Volatility + Relative-Value + Risk Control	ML predicts IV/RV/returns → finds mispricing → picks strategy → sizes by risk	Maximize net risk-adjusted alpha	5	4.5	Strongest academic support: RFS study, 12M+ observations 1996–2020, ML option-return portfolios profitable after costs
2	ML Multi-Strategy Regime-Switching	Detect regime → activate the strategy that fits it	Consistency across regimes	4.5	5	Option payoffs depend on regime; AQR found buying options only paid in the top decile of volatility changes
3	ML + Deep Hedging / RL	ML finds opportunity; RL decides entry, size, hedge, exit	Maximize net P&L after costs and risk	5	5	Static IV fit and real hedging results diverge once costs and risk aversion are included
4	Institutional VRP / Risk-Target	Systematically sell volatility with vol targeting and tail controls	Stable VRP harvesting	4	4.5	Mirrors public AQR, Parametric / Morgan Stanley and Man Group approaches
5	Institutional Option-Overlay / Outcome	Options change a portfolio's risk/return (puts, covered calls, collars)	Optimize portfolio outcome	3	5	Mirrors BlackRock and Morgan Stanley overlay and income strategies

Framework details

- **#1 targets:** IV change, realized vol, IV-RV spread, expected option return, P(profit > 0), CVaR. Models: XGBoost (tabular), LSTM (sequences), Transformer (long-range), GNN/CNN (surface),

Elastic Net (benchmark), meta-model (combine). Objective: maximize Sharpe + Sortino minus penalties for CVaR, max drawdown and turnover, within Greek and position limits.

- **#2 regimes:** low IV / low RV, high IV / falling RV, high IV / rising RV, bull, bear, crisis, earnings, liquidity stress. Example rules: high IV + stable RV → short-vol spreads; high IV + rising RV → cut short vol; extreme VIX → defined risk only; illiquid → no trade. Supported by the IMF 2026 GFSR on dealer gamma and 0DTE growth.
- **#3 reward:** P&L minus costs, CVaR, drawdown and turnover penalties — not raw P&L. RL is the decision engine, not the alpha engine.
- **#4 caution:** AQR warns sharp moves create meaningful VRP drawdowns, so sizing must reflect tail risk.
- **#5 goal:** reduce drawdown and volatility, generate income, protect concentrated positions — not maximum profit.

Scorecard highlights (5 = best): frameworks 2 and 3 score 5 on drawdown and CVaR control; #1 scores 5 on profit and Sharpe; #4 and #5 score highest on interpretability and institutional resemblance; #3 has the lowest interpretability and highest cost sensitivity.

Transaction-cost warning: a 2026 LSTM study saw raw profits eroded by bid/ask, with delta-hedged results turning negative at large spreads. The agreed rule: trade only if expected edge > bid/ask + slippage + commission + market impact; otherwise no trade.

Discussion 4 — Framework × horizon and the objective hierarchy

Daily plus medium-term was chosen as the primary research horizon, with intraday as a separate later module. ML-heavy frameworks fit daily and medium term best; institutional frameworks gain ground at long horizons.

Framework	Main goal	Intraday	Daily	Medium	Long
1 Adaptive ML	Net risk-adjusted alpha	4.5	5	5	4
2 Regime-Switching ML	Consistency across regimes	4.5	5	5	5
3 ML + Deep Hedging / RL	Dynamic P&L after costs	4	5	5	4.5
4 Institutional VRP	Stable VRP harvesting	3	4.5	5	5
5 Institutional Overlay	Portfolio outcome	2	4	4.5	5

- **Intraday:** edge must beat bid/ask, slippage, commissions, impact and latency; favour fast models and a strict trade filter.
- **Daily:** best compromise. A July 2026 Federal Reserve paper using boosted trees on moneyness/maturity partitions cut one-month-ahead IV RMSE by 13% versus its benchmark.
- **Medium term (1 week–3 months):** Regime ML ranks first; more time to overcome costs.
- **Long term (3 months+):** Regime ML, Institutional VRP and Overlay lead; structural risk premia matter more than short-term signals.

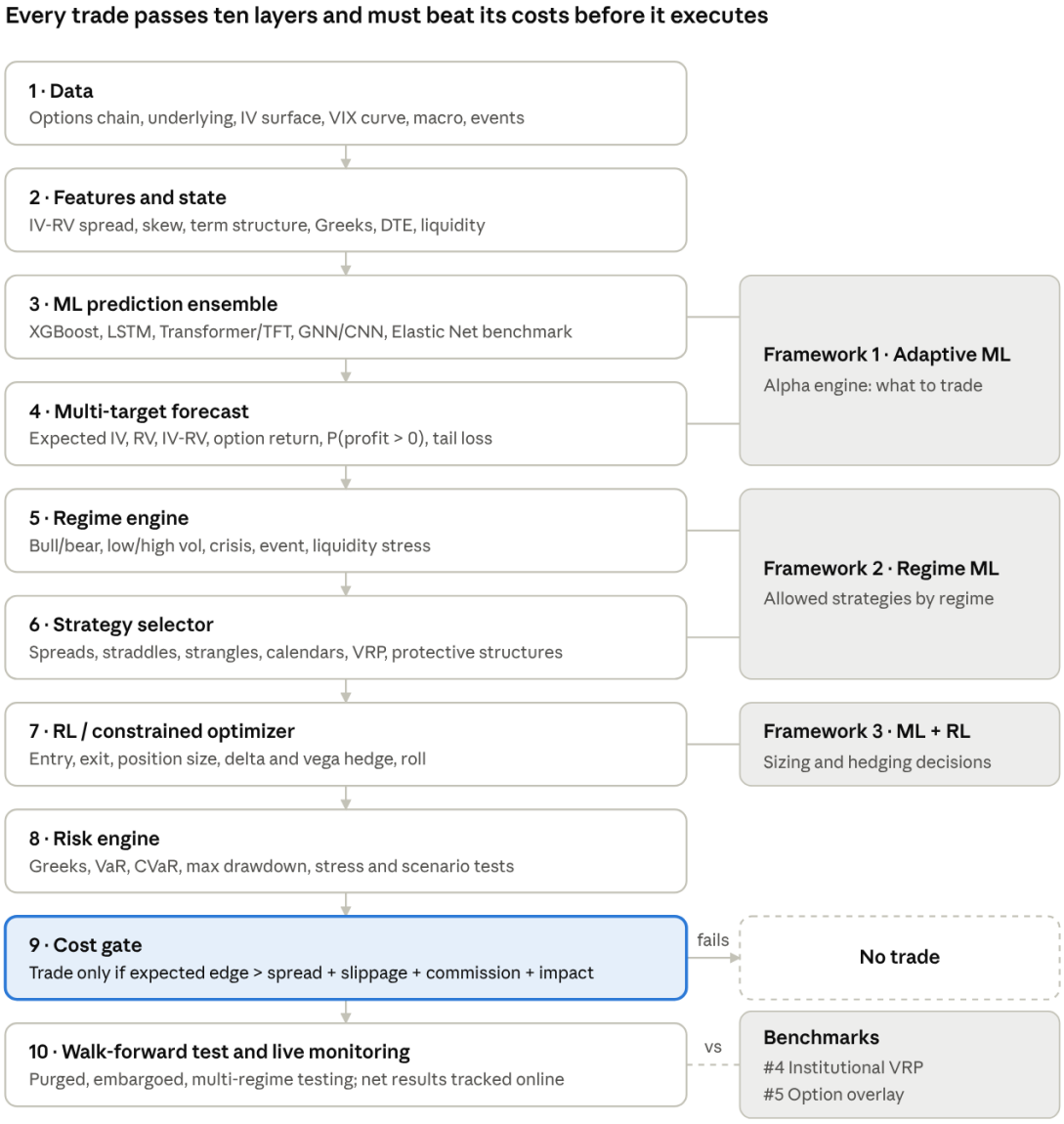
Objective hierarchy agreed for the project

Priority	Objective	What is measured
1	Survival	CVaR, stress loss, probability of ruin
2	Risk control	Max drawdown, Greeks, concentration, liquidity
3	Risk-adjusted profit	Sharpe, Sortino
4	Net profit	CAGR / total P&L after all costs
5	Efficiency	Return per unit of turnover, margin, capital
6	Prediction quality	RMSE, MAE, AUC — supporting metrics only

Ultimate objective: maximum sustainable net risk-adjusted return, subject to predefined limits on CVaR, max drawdown, Greeks, turnover, liquidity and margin.

Recommended final architecture

The end state is a hybrid of frameworks 1, 2 and 3, validated against 4 and 5, where prediction, strategy choice, sizing, risk and execution are separate layers rather than one model deciding everything.



recommended ML options system · 10 layers, 3 frameworks, 2 benchmarks

Signals flow top to bottom; any trade whose expected edge does not clear total costs stops at the cost gate.

Decisions carried forward

- Core model families: XGBoost, LSTM, Transformer/TFT, GNN (or CNN), PPO/deep hedging, with Elastic Net as the control.
- Prediction layer and decision layer stay separate.
- Optimize net P&L after realistic costs, not RMSE.
- Primary horizon: daily + medium term; intraday as a separate later module.
- Institutional VRP and option-overlay frameworks act as benchmarks the ML system must beat.

- Planned next study: top 20 option strategies × top 20 ML models × features × DTE × delta × IV/RV × VIX regime × costs, tested walk-forward on Sharpe, Sortino, max drawdown, CVaR and probability of profit.

Critical review

The conversation's core logic is sound and matches mainstream quant practice, but its rankings are subjective, several numbers conflict internally, and many cited 2026 studies should be verified before they drive decisions.

Strengths

- Puts after-cost, out-of-sample economics ahead of prediction accuracy — the right priority.
- Strong emphasis on leakage, walk-forward testing, crisis tests and calibration.
- Separating prediction, strategy selection, sizing, risk and execution is a defensible design.
- Keeps simple benchmarks (Elastic Net, institutional VRP) that complex models must beat.
- Cites peer-reviewed and institutional sources (RFS, Gu–Kelly–Xiu, AQR, BIS, IMF, Bank of England).

Internal inconsistencies to resolve

Issue	Where it appears	Why it matters
Bi-LSTM called "best", yet its quoted MSE (4.65%) is worse than plain LSTM (4.35%) and ConvLSTM (4.44%)	Discussion 1, model #2	Either the figures or the claim is wrong; check the source paper
Model order shifts: Transformer #3 and RF #7 overall, but LightGBM #3 and RF #5 in the tiered list	Discussion 1	The ranking is opinion, not a stable result
Transformer marked green (suitable) for intraday in one table but 3/5 in the matrix	Discussion 2	Use the matrix; state the latency caveat
PPO is in the "top five to build", yet also "not the first model"	Discussions 1 and 3	Sequence it: build prediction and risk layers first, RL later
Nearly every model and framework scores 4–5 stars	All tables	Ratings do not discriminate; treat them as hypotheses to test
Success-factor scores (9.9 vs 9.8) imply false precision	Discussion 1b	Use tiers (critical / important / useful) instead

Claims that need verification

- Many cited items are 2026 papers, SSRN preprints or arXiv drafts; confirm each exists and says what is claimed before relying on it.
- Specific numbers to re-check: 18.2M minute-bar observations; LSTM R^2 ~53%; GNN ≥8.81% RMSE gain; 13% Federal Reserve IV RMSE reduction; RFS 12M+ observations study.
- Gu–Kelly–Xiu figures (Sharpe 0.77 vs 0.51; 1.35 long-short) come from US equities, not options; transfer to options is an assumption.
- The Sharpe/RMSE worked examples (e.g. Elastic Net 0.72 vs XGBoost 0.81) are illustrations, not results.

Gaps not covered

- Data sourcing and cost: which vendor (e.g. OptionMetrics, ORATS, CBOE DataShop), history length, and budget.
- Compute and infrastructure plan, especially for Transformers, GNNs and RL.
- Statistical safeguards against multiple testing (e.g. deflated Sharpe, White's reality check) — critical for a 20 × 20 grid.
- Margin, capital requirements and early assignment for American-style options.
- Concrete regime definitions (thresholds for VIX, IV/RV, term structure).
- The earlier "top 20 option strategies" study is referenced but not included here.
- Nothing guarantees future profitability; ChatGPT itself notes no cited study proves a live strategy will stay profitable.

Next steps

Build the simplest version end to end first, then add complexity only where it beats the benchmark after costs.

1. Verify the key papers and numbers flagged in the review.
2. Bring in the "top 20 option strategies" study so the strategy × model grid can be defined.
3. Choose a data vendor and lock the dataset: underlyings, date range, frequency, bid/ask quality.
4. Define targets (IV change, RV, IV-RV, P(P&L > 0), expected P&L) and write down regime thresholds.
5. Build baselines: Elastic Net, institutional VRP rule, simple overlay.
6. Add XGBoost/LightGBM, then LSTM; test walk-forward with realistic costs and the cost gate.
7. Add regime engine and strategy selector; then Transformer/GNN; then RL for sizing and hedging.
8. Apply multiple-testing corrections before calling any result an edge; paper-trade before any capital.

Sources

Source conversation: [Project Option trading — shared ChatGPT chat]. The references below are those ChatGPT cited; they were not independently opened for this review.

- **Option-specific ML:** Predicting option prices from their price history (Review of Derivatives Research, 2026); Option Return Predictability with Machine Learning and Big Data (Review of

Financial Studies); Option pricing using Machine Learning (Expert Systems with Applications); Non-parametric option pricing before and during COVID-19; Forecasting implied volatilities of currency options with ML; GNN momentum spillover in option pricing (Journal of Futures Markets); CNN implied-volatility surface prediction; VAE option pricing on NIFTY50; Neural-SDE pricing and hedging; Gaussian-process and deep-kernel American option pricing; Learning sequential option hedging models (Journal of Banking & Finance); Deep RL algorithms for option hedging; Hedging American puts with deep RL; When Lower Pricing Error Trades Worse (SSRN, 2026).

- **General financial ML:** Gu, Kelly & Xiu, Empirical Asset Pricing via Machine Learning (RFS / NBER / AQR); DNN, gradient-boosted trees and random forests for S&P 500 statistical arbitrage; Transformers vs simple NNs for stock returns; ML stock-market volatility forecasting; Office of Financial Research 2026 forecasting benchmark; Federal Reserve July 2026 tree-based IV forecasting.
- **Institutions:** Bank of England Financial Stability Report (July 2026); BIS papers on ML for market monitoring and the expected-shortfall framework; IMF Global Financial Stability Report 2026 and AI-in-markets papers; AQR on the volatility risk premium and buying options; BlackRock on options and overlays; Morgan Stanley / Parametric VRP and Global Balanced Income; Man Group risk premia; MSCI on nonlinear factor effects; CFA Institute on deep learning and trade execution.